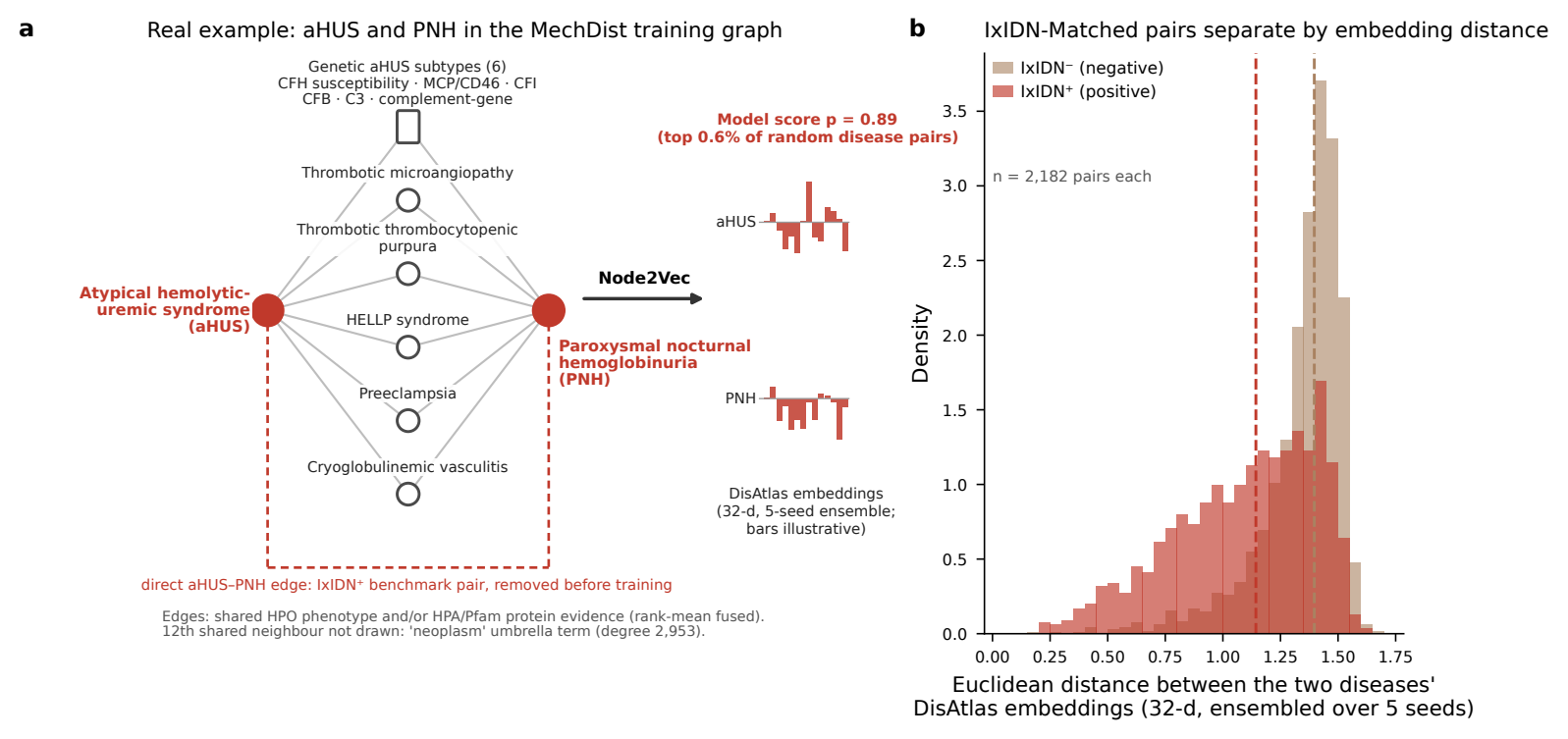
a
Real example: aHUS and PNH in the MechDist training graph
Genetic aHUS subtypes (6)
CFH susceptibility · MCP/CD46 · CFI
CFB · C3 · complement-gene
Thrombotic microangiopathy
Thrombotic thrombocytopenic purpura
Atypical hemolytic-uremic syndrome (aHUS)
HELLP syndrome
Preeclampsia
Cryoglobulinemic vasculitis
Paroxysmal nocturnal hemoglobinuria (PNH)
Node2Vec
Model score p = 0.89
(top 0.6% of random disease pairs)
aHUS
PNH
DisAtlas embeddings
(32-d, 5-seed ensemble;
bars illustrative)
direct aHUS–PNH edge: IxIDN+ benchmark pair, removed before training
Edges: shared HPO phenotype and/or HPA/Pfam protein evidence (rank-mean fused).
12th shared neighbour not drawn: 'neoplasm' umbrella term (degree 2,953).
b
IxIDN-Matched pairs separate by embedding distance
IxIDN⁻ (negative)
IxIDN⁺ (positive)
n = 2,182 pairs each
Density
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Euclidean distance between the two diseases'
DisAtlas embeddings (32-d, ensembled over 5 seeds)